LESSON PLAN

AIRCRAFT WIRING PRACTICES (ELECTRICAL WIRING AND INTERFACE SYSTEM) (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects.
5. Specific Objectives	12.1.6.1 Trainee will be able to understand hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re-circulation effects.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight dynamics diagrams; Instructor supervision of diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight phases covered in this lesson?
(b) Define the term 'free air hover'.
(c) List the aerodynamic effects occurring during vertical descent.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Hovering and forward flight principles • Take off and climb to free air hover • Vertical descent dynamics	L & Ds	PPT, IB	15 min
(b)	• Ground effect aerodynamics • Ground cushion effects • Re-circulation effects analysis	L & Ds	PPT, PDF	20 min
(c)	• Transition from hover to forward flight • Climb performance characteristics • Descent rate variations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between ground effect and ground cushion.
(b) Describe the re-circulation effects during vertical descent.
(c) How does forward flight differ from hovering in terms of lift generation?

12. Home Assignment and Follow Up	Prepare a brief note on the transition from LP 02 to this lesson regarding flight envelope limitations.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Auto rotation, ground resonance, vortex ring, blade sailing.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min